GENERATIVE AI FOR FINANCE

Week 1: Introduction

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What you will learn

One

How to work with AI to do financial analysis

Two

How custom chatbots and AI agents work

Three

How to work with AI to build custom chatbots and AI agents for financial analysis

> Item #1 is essential today. Item #2 is valuable today.

- More practice on #1
- A deeper understanding of #2
- You could end up at a small firm and be the AI expert

The purpose of #3

Course schedule

WEEK 1	Introduction
WEEK 2	Skills and plugins
WEEK 3	Local models and AI security
WEEK 4	Custom chatbots and apps
WEEK 5	Automating DCF analysis
WEEK 6	Deployment and databases

Four elements, 25% each. The first three are group projects.

One

AI cost of capital exercise: calculation, visualization, report generation

Two

Cost of capital app

Three

Valuation Skill for Claude

Four

Seated exam with AI allowed

> For each group project the deliverables are a chat, app, or agent, and a two-page Word doc explaining development and use.

Evaluation

HPE’s experience with AI agents

> “[Alfred] removed about 90% of the manual effort that once went into preparing our weekly financial review. We used to spend hundreds of hours preparing 100+ slides for a Monday meeting.”
>
> Marie Myers, CFO, Hewlett Packard Enterprise (Fortune, Feb 2026)

Today’s session

- How large language models work
- The universe of models
- Using Claude
- Effective use of AI

- Chatbots and agents
- Claude and Excel for valuation
- Corporate implementations of AI

HOW LARGE LANGUAGE MODELS WORK

Tokenization

Text is cut into pieces from a fixed vocabulary before the model sees any of it.

```
"The cat sat on the mat"  →  ["The", " cat", " sat", " on", " the", " mat"]

"Unbelievable"            →  ["Un", "believ", "able"]
```

Common words stay whole; rare and long words break into pieces. A typical vocabulary runs to about 100,000 tokens.

> Numbers break on the same rules, which have nothing to do with arithmetic. `13,963,378.65` reaches the model as a handful of unrelated pieces.

Vectors

Each token is associated with a vector, meaning a list of numbers. So, a sequence of text becomes a sequence of lists of numbers.

Each token is a list

“king” becomes a list of 4,096 numbers. “queen” becomes a different but somewhat similar list of 4,096 numbers.

Tokens used in similar contexts end up near each other.

The space is mostly empty

There are far more possible vectors than there are tokens, so most points in the space name nothing.

The prediction can land between tokens. The “closest” token is usually selected but not always (so LLM output is somewhat random).

The whole task

Given the tokens (lists of numbers) so far, predict the token (list of numbers) that comes next. That is the entire operation.

Everything a model appears to do — answering, summarizing, writing code, refusing — is that one prediction, repeated.



Words

Input text



Tokens

Text split into units



Vectors

4,096 numbers per token



Neural Network

Billions+ of parameters



Vector

Output vector



Token

Predicted next token

The prediction pipeline

A neural network is a function

THE IDEA

$y = 4x$ is a function. The 4 is a parameter.

Change it to 3 and you have the same structure with different behavior.

THE SCALE

Same idea with perhaps trillions of parameters, arranged as a transformer — the architecture published by Google researchers in 2017.

A model trained only to predict the next token continues your text. It does not answer a question.

1. Role tokens. The text is wrapped in markers: `◁user▷` your question `◁assistant▷`.
2. Instruction tuning. The model is further trained on examples where what follows `◁assistant▷` is a useful reply.
3. Reinforcement learning from human feedback. Raters score replies, and the model is trained toward what raters preferred.

The mechanism did not change. What changed is which continuation is the likely one. GPT became Chat-GPT in 2022.

From prediction to conversation

THE UNIVERSE OF MODELS

WHEN	WHAT
2017	Google researchers publish the transformer (“Attention Is All You Need”)
2020	GPT-3 (OpenAI) shows abilities that appear with scale
2022–2023	Online chatbots. You typed, it typed back
2023–2024	Code execution and web search arrive
2025	The serious versions move to the command line, reading and writing files on your own machine. DeepSeek R1 appears
2025–2026	That command-line capability is wrapped back into an app
2026	Agents that carry out extended work on your files without you living in a terminal

A brief history

The frontier: closed weights

PROVIDER	MODEL	RELEASED	INPUT / OUTPUT, $ PER M TOKENS
Anthropic	Claude Fable 5.1	Sep 2026	10 / 50
Anthropic	Claude Opus 5	Jul 2026	5 / 25
Anthropic	Claude Sonnet 5	Jun 2026	2 / 10
OpenAI	GPT-6 Astra	Sep 2026	10 / 50
OpenAI	GPT-5.6 Sol	Jul 2026	5 / 30 (4 / 20 until Nov 21)
Google	Gemini 3.8 Flash	Sep 2026	1.50 / 7.50 (0.75 / 3.75 until Dec 31)
xAI	Grok 4.6	Aug 2026	2 / 6
Meta	Muse Spark 1.3	Sep 2026	1.25 / 4.25

Prices are as of September 18, 2026.

PROVIDER	MODEL	RELEASED	INPUT / OUTPUT, $ PER M TOKENS
Alibaba	Qwen3.8-Max	Aug 2026	2 / 6
Zhipu (Z.ai)	GLM-5.3	Aug 2026	1.40 / 4.40
Moonshot	Kimi K3	Jul 2026	3 / 15
DeepSeek	DeepSeek V4 Pro	2026	1.32 / 3.96
NVIDIA	Nemotron 3 Ultra	Jun 2026	free
Mistral	Mistral Medium 3.5	Apr 2026	1.50 / 7.50

> Open weights means anyone can download the model and run it on their own hardware. The prices are for the developer’s hosting; DeepSeek charges half off-peak.

The frontier: open weights

An aggregation across multiple benchmarks

Artificial Analysis Intelligence Index, v4.3, best setting for each model.

MODEL	SCORE	MODEL	SCORE
Claude Fable 5.1	53	GLM-5.3 (open)	45
GPT-6 Astra	53	Qwen3.8-Max	45
Claude Opus 5	51	Kimi K3 (open)	44
Muse Spark 1.3	48	Grok 4.6	44
GPT-5.6 Sol	47	Gemini 3.8 Flash	41

> The best open model trails the best closed ones by 8 points.

Cost per task is the price per token times the number of tokens the model uses.

MODEL, EFFORT SETTING	INDEX SCORE	COST PER TASK
GPT-6 Astra, max	53	$3.26
Claude Fable 5.1, max	53	$7.63
Claude Opus 5, max	51	$5.86
GPT-5.6 Sol, max	47	$1.99
GPT-6 Astra, low	46	$0.82

> Astra and Fable 5.1 have the same list price, $10 / $50. At max effort Astra writes about 27,000 output tokens per task and Fable 5.1 about 78,000. Artificial Analysis, September 2026.

Cost per task

Enterprise adoption

	ANTHROPIC	OPENAI	GOOGLE	OPEN SOURCE
Enterprise LLM spend (Menlo, Dec 2025)	**40%**	27%	21%	12%
Company usage (Datadog, Apr 2026)	~46% (+23 pts YoY)	**63%** of companies	~37% (+20 pts YoY)	13% of workloads
Businesses paying (Ramp, Aug 2026)	**43.8%**	39.8%	?	3.6%
Code generation (Menlo, Dec 2025)	**54%**	21%	?	?

Anthropic is #1 by enterprise spend; OpenAI is #1 by breadth of adoption. The default enterprise strategy is **multi-model** — 69% of companies use 3+ models, routing different tasks to different providers. Closed-source models power 85–90% of enterprise workloads. The Ramp row is the share of 70,000+ US businesses paying each provider, and many pay more than one.

Coding

Claude Opus and Sonnet are the best coding LLMs, and most of what we do in weeks 2 through 6 is code.

Excel

Anthropic released Excel integration, so Claude can generate fully functioning, nicely formatted workbooks.

Agents

Anthropic created the Model Context Protocol and Claude Skills, which simplify building agents.

> OpenAI also uses MCP and has a coding agent along the same lines as Claude Code.

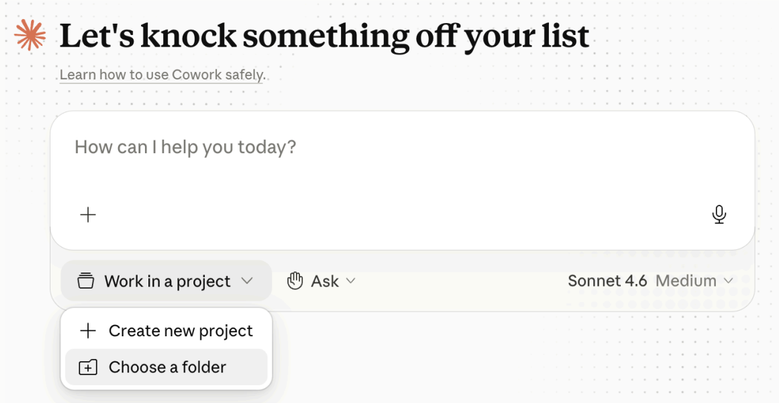
Why Claude for this course

USING CLAUDE

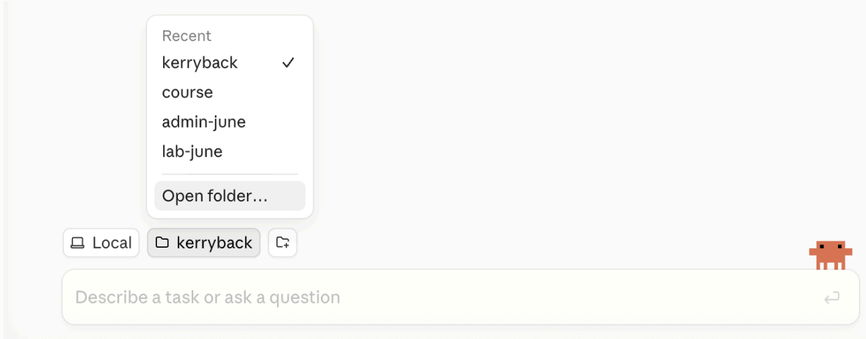
Project folder

In **Cowork** and **Code** modes, open a folder to give Claude access to your local files.

Cowork — click *Work in a project* at the bottom of the chat window, then *Choose a folder*



Code — click the folder icon in the toolbar or use the folder dropdown to select *Open folder...*



For **Code**, selecting a folder tells Claude where to look first — it can still work outside the folder. For **Cowork**, the folder determines where the virtual machine is mounted — nothing outside it is visible or can be touched.

What Claude can reach

Claude Code

The command line, and Claude Desktop → Code → Local

Can reach anywhere on your hard drive, but looks first in the project folder

Claude Cowork

Can reach only the project folder

Claude Chat

`claude.ai`

You must upload files

> We install Claude Code in week 2. This week is Claude Desktop and the Excel add-in.

Permissions

Cowork

Selector in the chat box:

- **Manually approve** — Allow or Deny each action
- **Automatically approve** — Claude safety-checks each action and blocks unsafe ones
- **Skip all approvals** — nothing is checked

Computer use always requires approval

Code — Local

Selector by send button

- **Manual** — asks before every edit and command
- **Accept edits** — edits go through, commands still ask
- **Plan** — reads and proposes, changes nothing
- **Auto** — a classifier approves, blocking risky actions
- **Bypass permissions** — no prompts, no checks

Code — Cloud

Selector by send button

- **Accept edits** — edits go through, commands still ask
- **Plan** — reads and proposes, changes nothing
- **Auto** — a classifier approves, blocking risky actions

EFFECTIVE USE OF AI

Start a new conversation for a new task

AI reads the entire conversation (the context window) each time you prompt. Irrelevant stuff consumes tokens and may be confusing.

When a conversation gets long

Tell AI to write a handoff document so you can start a fresh conversation. How long? There is a tradeoff; it depends on how much is irrelevant.

Managing conversations

Good enough prompting

Most ‘prompt engineering’ techniques from a few years ago are now either useless or actually counterproductive. Don’t tell it “you are an expert data scientist” or “think step by step.”

Tell it what you want, not what to do

I want ... How can we do it?

Ask AI for help in prompting

If you are unsure how to ask something, ask AI how to ask it.

If something doesn’t work

Try again.

> Treat AI as an infinitely patient colleague with a poor memory.

Your go-to prompt

Think about the best way to do this, and let's develop a plan together.

Ex-post diagnosis

Prompt: Explain what you did step by step. Did you make any decisions that we have not discussed? Write this as an HTML doc.

Use AI to evaluate

Start a new conversation or ask for a subagent. Ask it to evaluate the first agent's work, summarized in the HTML doc, relative to your goals.

Plan, evaluate, critique

What this buys

1. AI is a prediction machine. Ask it to do something, and it will probably do what it has seen most often in its training data.
2. Ask it the best way to do something, and it will regurgitate evaluations of methods it has seen.
3. Ask it to proceed after explaining the best way, it will read the “the best way is …” in the conversation history and follow that.

4. Misunderstandings and unstated assumptions are the most common sources of errors. Discuss before and after to surface those.
5. AI is not good at critiquing itself in the same conversation, because it reads its reasoning in the conversation history and predicts the same thing. You need a fresh context window to get a reliable critique.

Numbers need code

If a number appears and no code produced it, it was predicted. Do this instead.

> **Prompt**
>
> `Compute this in Python. Do not report a number you did not compute with code.`

Asked to sum three numbers 83, 16, 104, AI is much better at predicting the code `sum([83, 16, 104])` than it is at predicting what should follow `83 + 16 + 104 =`

AI predicts text. It does not check what it produces against a source.

Hallucinations are most likely when it is asked to recall something specific that is not in front of it:

- Citations, quotes, and URLs
- Numbers about a particular company, year, or dataset
- Options and arguments of niche or recent software

Verify the facts

Ask for a source and look it up.

Hallucinations

Hallucinations in code

Caught by running it

A made-up function or argument raises an error as soon as the code runs. The agent reads the traceback and fixes it.

Not caught by running it

The wrong column, a join that drops rows, a filter on the wrong year. The code runs and returns a number.

> The second kind is a misunderstanding or an unstated assumption, not a hallucination. Discussing the plan beforehand and the decisions afterwards is what surfaces it.

CHATBOTS AND AGENTS

Chatbots and agents

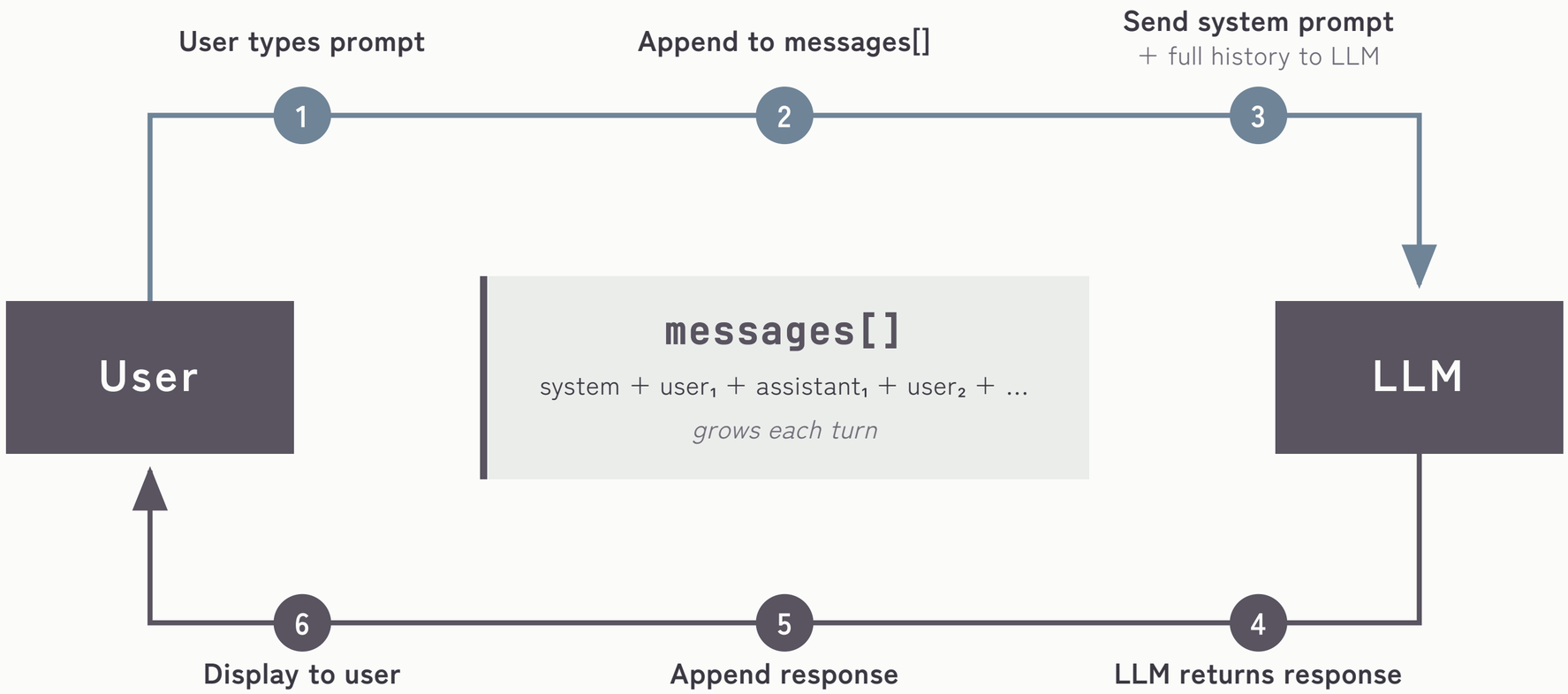
Chatbots

- System prompt with base instructions
- Maintain conversation history
- LLMs are stateless — no memory of prior prompts & responses

AI Agents

- Chatbots with tools
- The LLM writes instructions to use tools
- Examples: retrieve data, execute code, generate charts, Excel workbooks, Word docs, PowerPoint decks, images, videos

Anatomy of a chatbot

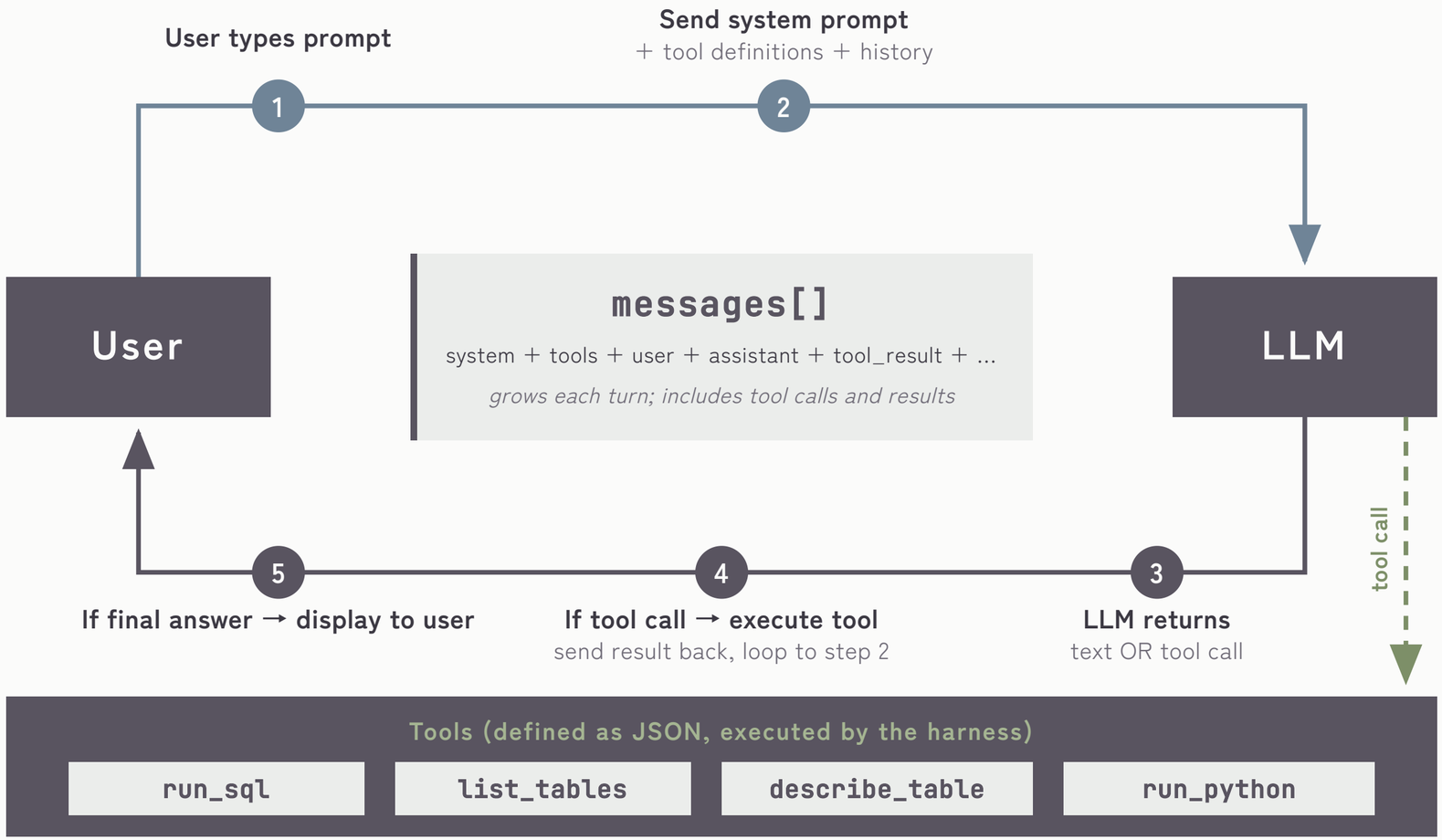


Anthropic’s system prompts

Anthropic publishes the system prompts for its models:

https://docs.claude.com/en/release-notes/system-prompts

Anatomy of an agent



The tools of Claude Code

TOOL	WHAT IT DOES
Read, Write, Edit	Open and change files — documents, spreadsheets, code
Bash	Run commands and programs on the machine
Glob, Grep	Find files, and search inside them
WebSearch, WebFetch	Run a search, then download and read a page
Task	Start a subagent with its own instructions
MCP connectors	Call an outside service: a database, an API

CLAUDE AND EXCEL

How Claude builds a workbook

Creating a workbook

Claude writes Python — the openpyxl library — that loops over every cell and inserts data, formulas, and formatting.

Editing a workbook

To edit a workbook, Claude reads the XML and edits it. An `.xlsx` unzips into one XML file per worksheet, with the text in `sharedStrings.xml` and the formatting in `styles.xml`; each cell is an element holding a value or a formula. Claude rewrites those elements and repacks the file, so charts, pivot tables, and formatting it did not touch are unchanged.

The formulas are live. A workbook Claude builds opens in Excel as a model you can change.

The Claude plugin for Excel

A chat panel inside Excel itself — no folder, no Python, no terminal. It works on the sheet that is already open.

Installing

- In Excel, add it from `Home → Add-ins`
- If your organization blocks the Office Store, IT deploys it instead through the Microsoft 365 Admin Center
- Sign in to your Claude account when prompted

Example

	A	B	C	D	E	F	G	H	I	J	K
1	**Mortgage Amortization Schedule**										
2											
3	**Inputs**										
4	Loan Amount	$400,000									
5	Annual Interes	6.500%									
6	Term (Years)	30									
7	Payments per	12									
8	Start Date	01/01/2026									
9											
10	**Outputs**										
11	Total Payment	360									
12	Periodic Rate	0.5417%									
13	Monthly Paym	**$2,528.27**									
14	Total Paid	$910,177.95									
15	Total Interest	$510,177.95									
16											
17	**Amortization Schedule**										
18	**Period**	**Date**	**Beginning Balance**	**Payment**	**Principal**	**Interest**	**Ending Balance**	**Cumulative Interest**			
19	1	01/01/2026	$400,000.00	$2,528.27	$361.61	$2,166.67	$399,638.39	$2,166.67			
20	2	02/01/2026	$399,638.39	$2,528.27	$363.56	$2,164.71	$399,274.83	$4,331.37			
21	3	03/01/2026	$399,274.83	$2,528.27	$365.53	$2,162.74	$398,909.30	$6,494.11			
22	4	04/01/2026	$398,909.30	$2,528.27	$367.51	$2,160.76	$398,541.78	$8,654.87			
23	5	05/01/2026	$398,541.78	$2,528.27	$369.50	$2,158.77	$398,172.28	$10,813.64			
24	6	06/01/2026	$398,172.28	$2,528.27	$371.51	$2,156.77	$397,800.77	$12,970.41			
25	7	07/01/2026	$397,800.77	$2,528.27	$373.52	$2,154.75	$397,427.26	$15,125.16			
26	8	08/01/2026	$397,427.26	$2,528.27	$375.54	$2,152.73	$397,051.71	$17,277.89			
27	9	09/01/2026	$397,051.71	$2,528.27	$377.58	$2,150.70	$396,674.14	$19,428.59			
28	10	10/01/2026	$396,674.14	$2,528.27	$379.62	$2,148.65	$396,294.52	$21,577.24			
29	11	11/01/2026	$396.294.52	$2.528.27	$381.68	$2.146.60	$395.912.84	$23.723.84			

Claude

create a mortgage amortization table. assume a 30-year mortgage. allow the other parameters to be inputs.

I'll build a mortgage amortization table on Sheet1 — an inputs block (loan amount, rate, etc.) feeding a 360-month schedule for the 30-year term.

Let me set up the inputs section first.

Build inputs block in A1:B8 >

Now the calculated summary outputs:

Build outputs block in A10:B15 >

Payment of $2,528.27/mo checks out. Now the schedule headers and the first row, then I'll fill down 360 periods.

Prompt

```
Build an Excel workbook that values a company by discounted cash flow. Put the
assumptions on their own sheet, drive every cash flow off them with live formulas, and
add a sensitivity table over the discount rate and the terminal growth rate.
```

Change an assumption afterwards. If the valuation does not move, the number was typed in rather than computed.

Demo: a valuation workbook

Case: Valuing Walmart 2010

Harvard Business Publishing, W11058

1. Work the valuation with Claude and Excel rather than by hand.
2. Ask it to explain every assumption it made that you did not give it.
3. Start a fresh conversation and ask a second Claude to critique the first one’s valuation.

> Step 3 is the plan-evaluate-critique pattern from earlier in the session.

CORPORATE IMPLEMENTATIONS OF AI

Case: Implementing AI at Deloitte

Harvard Business Publishing, HEC382

OTHER READING

- CNBC on JPMorgan Chase’s LLM Suite
- State of AI in Business 2025
- Generative AI Reshaping Teamwork and Expertise

1. What is your experience with AI?
2. What do you hope to do with AI?
3. Where is your company at in implementing AI?

When we come back: what was common among your group members on each topic, and how did you differ?

Discussion

Week 2: Skills and plugins

- CLAUDE.md: standing instructions for a project
- Skills for repeatable tasks
- Plugins and marketplaces
- The same job as a prompt or a skill

The course ahead

- Week 3: Local models and AI security
- Week 4: Custom chatbots and apps
- Week 5: Automating DCF analysis
- Week 6: Deployment and databases

Upcoming